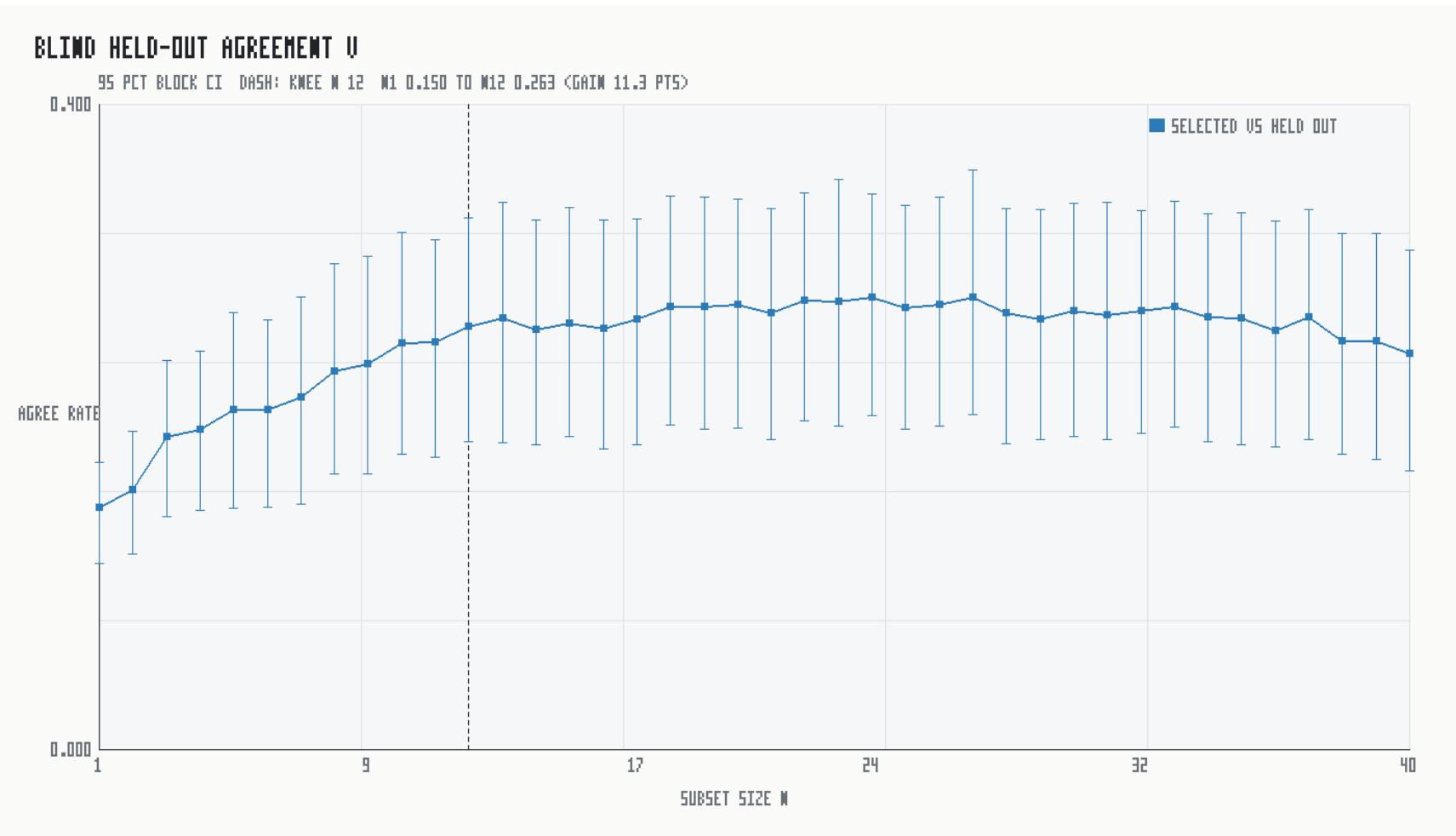
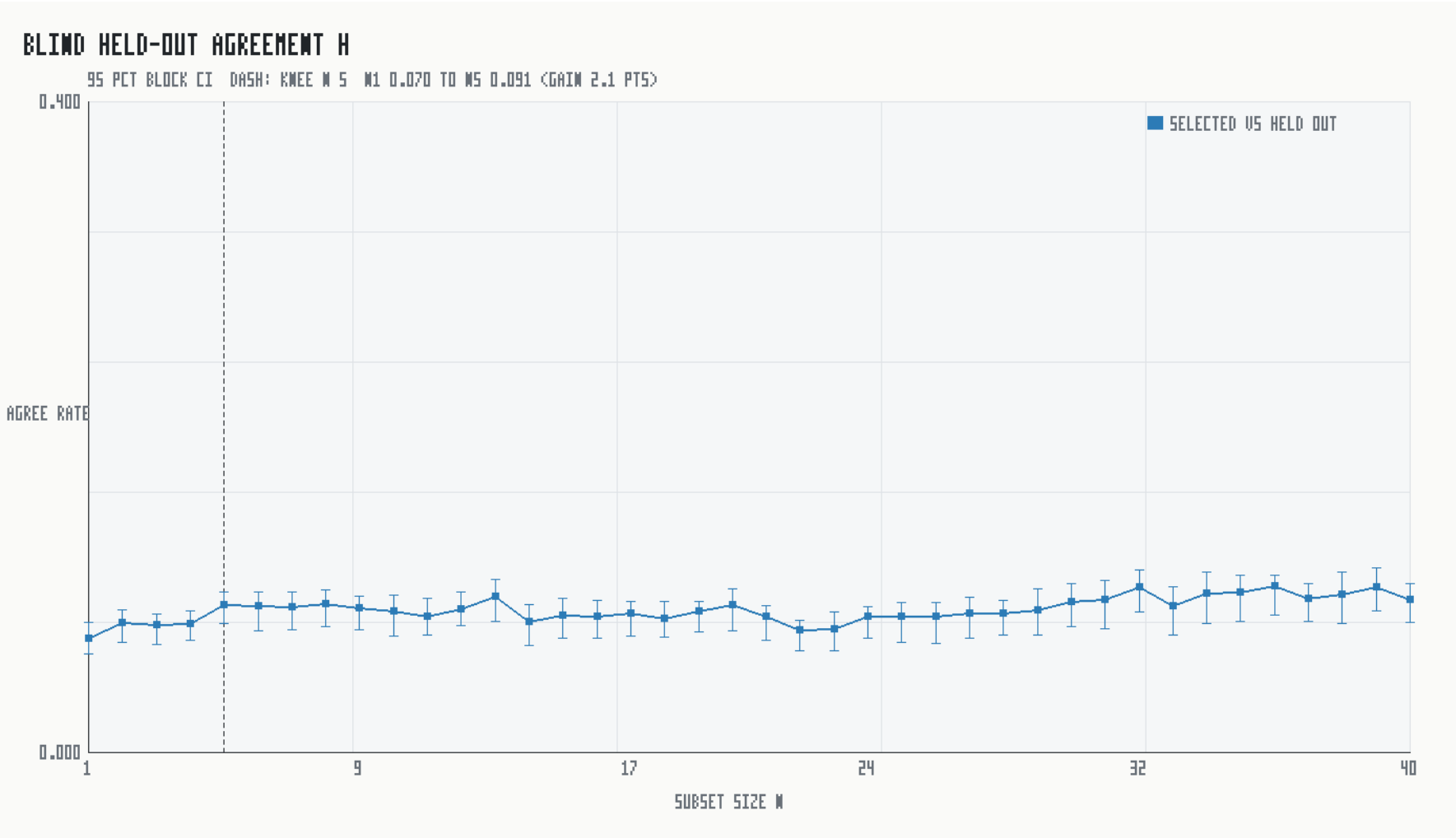


Turn-by-turn tune analysis using adaptive BPM ensembles in the Fermilab Mu2e Delivery Ring

Derek Steinkamp | Fermi National Accelerator Laboratory

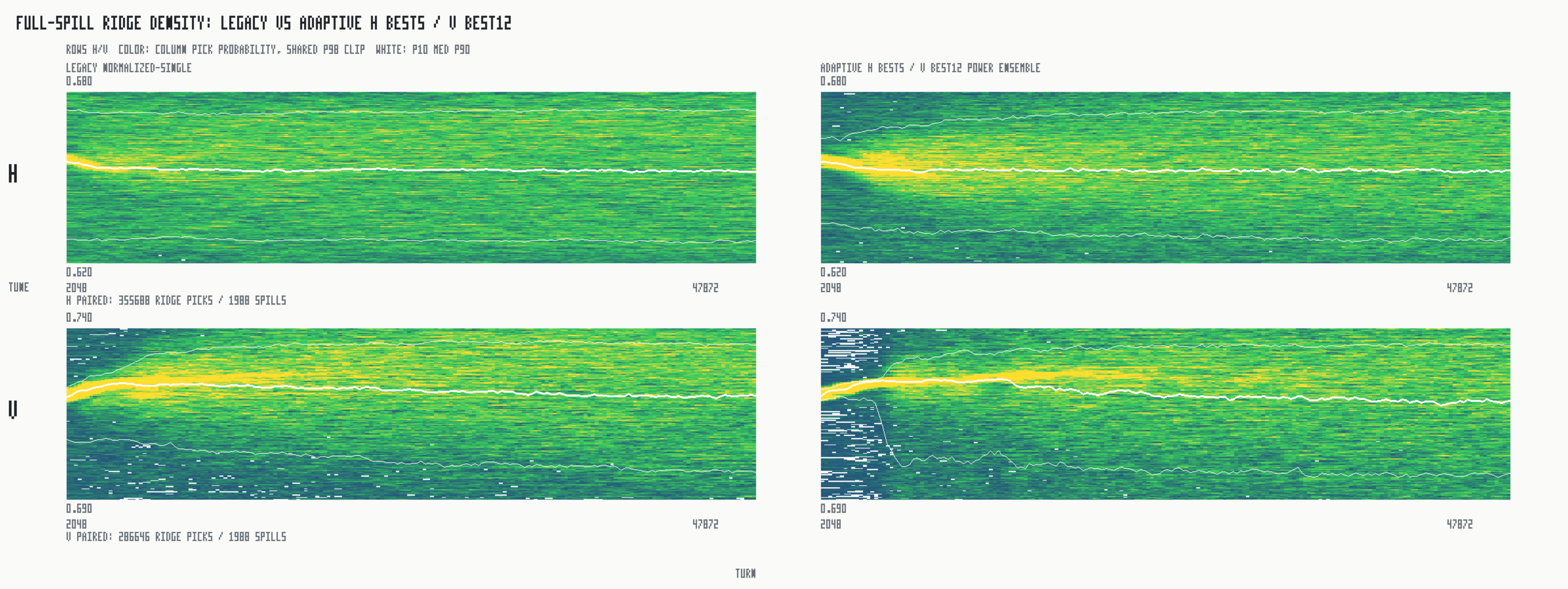
ADAPTIVE, LEAKAGE-CONTROLLED BPM ENSEMBLES

Synchronized 2,000-spill raw snapshots use a nominal 60 H / 60 V topology; 16 source absences across 12 flagged partial captures are hash-bound. Scaled threshold-substituted streams are excluded; members use eight fit windows with overlap purged. Later validation uses 1,000 stratified spill-plane cases across 5 held-out-digitizer folds.



H Best-5: blind full-band selected/held-out agreement 9.1% [7.9%, 9.8%]; median |Delta q| 0.0161. Reduced-sample knees span 2-13 in 5/7 runs; 2 unresolved.

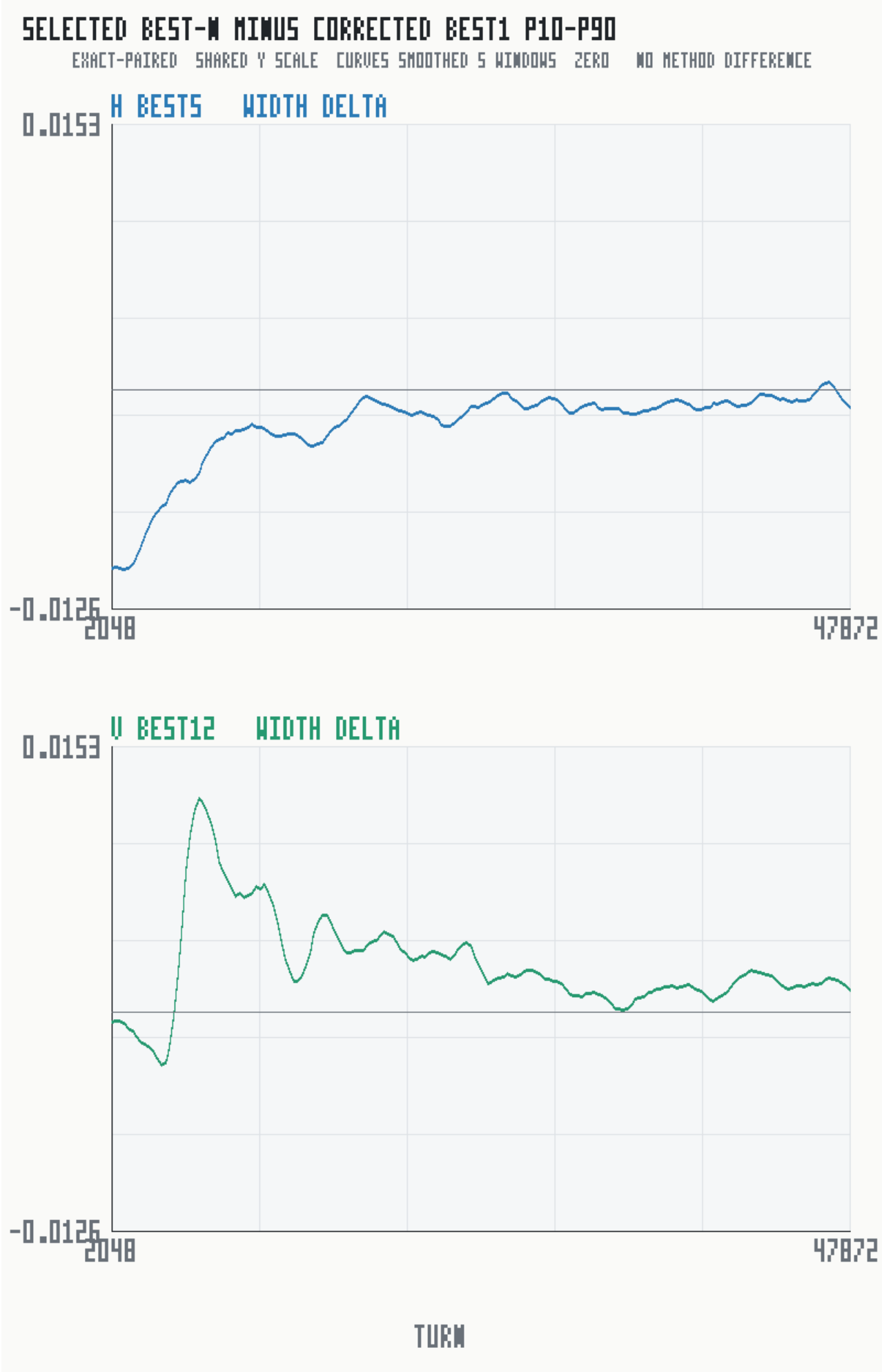
V Best-12: blind full-band selected/held-out agreement 26.3% [19.1%, 33.0%]; median |Delta q| 0.0107. Reduced-sample knees span 10-28 in 6/7 runs; 1 unresolved.



Exact-paired 50,000-turn density versus the audited legacy selector: **H Best-5** 355,688 picks; **V Best-12** 286,646. Legacy-to-selected change includes selector repair; color is pick probability, not power.

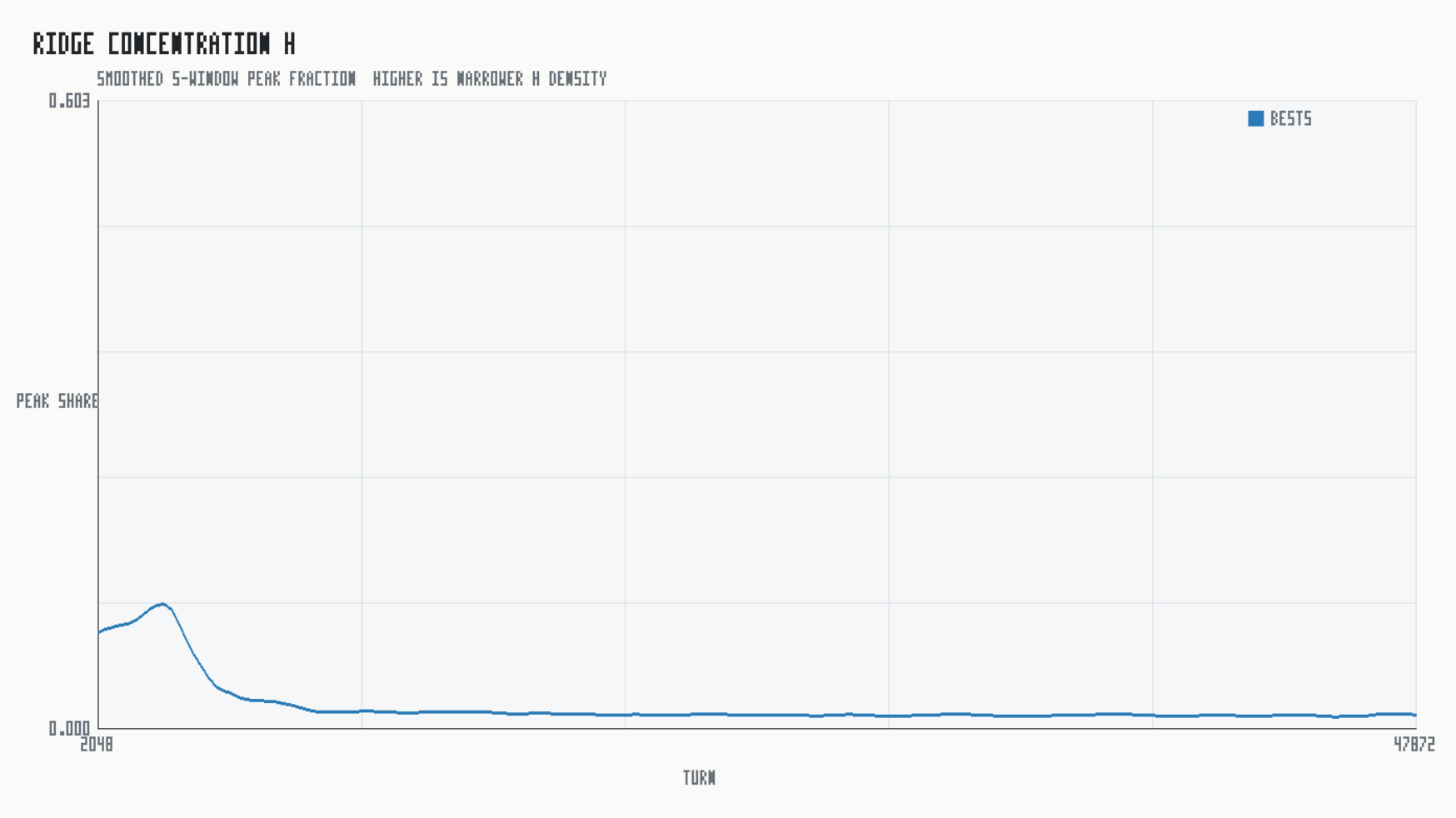
RESULT AND LIMIT

Versus adaptive Best-1, H Best-5 and V Best-12 improve blind agreement and median selected/held-out tune delta, with power-support tradeoffs. Full-buffer ensemble-size contrast: H narrower than corrected Best-1; V broader than corrected Best-1. Same-protocol all-training control favors Best-N in H 2/8 and V 3/8 comparisons; it favors all-training in H 3/8 and V 3/8. No external tune calibration is claimed.



Selected H Best-5 and V Best-12 P10-P90 width minus corrected adaptive Best-1 on exact paired spill/windows. Negative is narrower; zero is no ensemble-size difference.

2,000 spills; 4,000 H/V curve cases; 1,000 stratified validation cases; 5 held-out-digitizer folds. H Best-5 blind agreement 9.1%; V Best-12 26.3%. Median IQR change vs corrected Best-1: H -0.0024, V 0.001. Finite full-buffer pick coverage: H 99.7%, V 80.3%. Intensity weighting retained 0/240 tested effects.



Best-5 H concentration is tracked over 50,000 turns. Sustained half-peak loss begins near turn 5632. The multimetric change candidate is turn 5760. No extraction timestamp or cause is inferred.